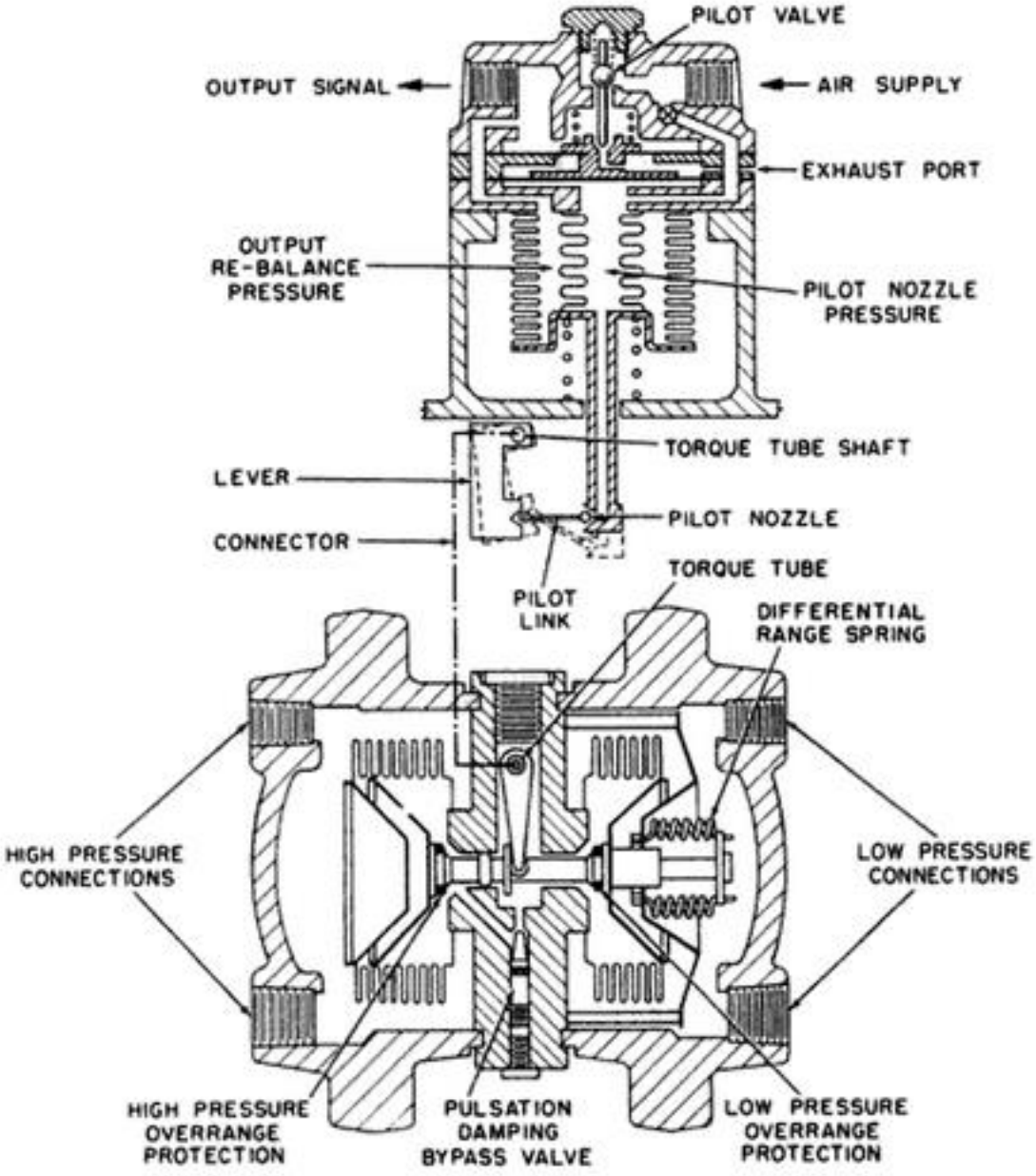


FIG. 2.11l
Linear flowmeter consisting of a combination of differential meter and square-root-extracting motion transmitter.

Square-Root-Extracting d/p Transmitter

There are differential pressure elements such as in Figure 2.11i that provide a motion output. A motion transmitter as in Figure 2.11g could be used with such a meter to give an output linearly proportional to the differential. However, the motion transmitter shown in Figure 2.11i converts linear motion of the meter into a square-root-related output so that orifice-type flow can be read on a linear scale. Linear signals are preferred when flows are added, subtracted, or averaged and when other analog computing and characterizing requirements exist. They are often specified in order to give better readability and control rangeability.

The meter in Figure 2.11i consists of a high- and a low-pressure bellows joined by a common center shaft. The differential causes the bellows to move a linearly proportional amount depending upon the total spring rate of the range spring, the bellows, and torque tube. Movement of the bellows twists the torque tube, causing rotation of the torque tube shaft. It is this motion that actuates the transmitter. The bellows are filled with liquid (usually ethylene glycol), and as the bellows move, liquid is transferred from one bellows to the other via the pulsation dampener needle valve. If the normal differential pressure range is exceeded, the bellows move until an O-ring on the center shaft seals off the liquid in the bellows. The pressure can then build up to the body rating of the meter without damaging the unit.

The torque tube rotates the lever connector of the motion transmitter. A floating pilot link is socketed in the connector at one end and baffles off the detector nozzle at the other. The transmitter is calibrated so that at zero differential the output is at 3 PSIG (0.2 bar) and the pilot link is horizontal. An increase in differential rotates the connector counterclockwise. This restricts the nozzle and causes a buildup in transmitted pressure, which in turn acts on the large bellows, moving it downward until equilibrium is attained. Since the pilot link has a fixed length and nozzle clearance is for all practical purposes constant, the pilot nozzle must be moved downward according to the cosine law, i.e., at minimum input with link horizontal, there must be considerable motion of the pilot nozzle as compared with the lever connector, and therefore the gain is practically infinite. As the included angle increases, considerably less nozzle travel is required to offset lever connector motion. For small angles, the relationship is almost exactly square root.

Another transmitter having the same function employs a varying spring rate with travel. This is accomplished by picking up added leaf springs as the travel increases.

Variable Area Flow Transmitter

When a variable area flowmeter is used as a primary flow-measuring device, a magnetic means of float position detection is utilized. The motion of the magnetic follower mechanism then is converted into a 3 to 15 PSIG (0.2 to 1.0 bar) signal. (Section 2.25 in the *Process Measurement* volume contains a detailed discussion of rotameters.)

In Figure 2.11j, a permanent magnet is embedded in either the float or in an extension of the float. A magnetic steel helix is supported in an aluminum cylinder mounted between bearings. The leading edge of the helix is constantly attracted to the magnet. The vertical position of the float results in a corresponding radial position of the helix. Attached to the helix follower assembly is a cam. The profile of the cam is sensed by a pneumatic detector circuit consisting of a transmitting and a receiving nozzle. The receiving nozzle pressure serves the same function as nozzle backpressure in a conventional baffle-nozzle circuit. When the flow from the transmitting to receiving nozzle is not interrupted by the cam, the receiver pressure is a maximum. When fully interrupted, receiver pressure is 0 PSIG (Pa). At balance, the detector system is throttled by following the cam profile.

As the float rises, the cam rotates in a direction of decreasing displacement. As the cam edge moves away from the detector nozzles, the increasing receiver nozzle pressure acts on the relay and results in an amplified increase in transmitted pressure. The transmitted pressure acts on the feedback capsule and moves the flexure-mounted detector nozzles toward the cam edge until equilibrium is established. The spring rate of the range spring is set so that a 12 PSI (0.8 bar) change in output is required to track the full displacement of the cam.

Span is adjusted by turning a screw that takes up coils in the range spring, thus changing its spring rate. Zero is set by adjusting a second screw that sets the initial spring tension.